23A01604b • OPEN CHANNEL FLOW

UNIT – 5

Flow Measurements and Hydraulic Modeling

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Sharp-Crested Weirs, Broad-Crested Weirs, Critical Depth Flumes
- Recent Advancements in Open Channel Flow Measurements
- Physical Modelling in Hydraulics
- Dimensional Analysis
- Modelling Closed Flows and Free Surface Flows
- Distorted Models
- Design of Physical Models

1. FLOW MEASUREMENT IN OPEN CHANNELS – OVERVIEW

1.1 Why measure flow?

Discharge data are needed for water allocation and billing in irrigation canals, flood forecasting, design of hydraulic structures, reservoir operation, environmental flow assessment and calibration of hydraulic models.

1.2 Methods of measuring discharge

Method	Principle	Typical use
Velocity–area	$Q = \Sigma v_i a_i$ from current-meter readings	Rivers and large canals
Hydraulic structures	Unique head–discharge relation (weirs, flumes)	Canals, labs, small streams
Dilution (tracer) gauging	Dilution of injected tracer: $Q = q(C_1 - C_2)/(C_2 - C_0)$	Turbulent mountain streams
Slope–area	Manning's equation using surveyed slope and section	Post-flood peak estimates
Acoustic / electromagnetic	Doppler shift, transit-time, induced voltage	Continuous, automated stations
Float / surface-velocity methods	Float time × surface-to-mean factor (0.85)	Rough estimates

1.3 Velocity–area method in brief

The cross-section is divided into verticals (spacing about 1/20 to 1/15 of width). At each vertical the mean velocity is taken as velocity at 0.6 d (one-point method), or the average at 0.2 d and 0.8 d (two-point method), or the average of 0.2, 0.6 and 0.8 d. Partial discharge $q_i = v_i \times d_i \times w_i$; total $Q = \Sigma q_i$ (mid-section method). A **rating curve** $Q = C\,(h - h_0)^b$ is then developed at the gauging station, where h is the stage and h_0 the gauge height at zero flow.

1.4 Hydraulic-structure approach

The idea is to force the flow through a **control section** where the flow passes critical depth (or a defined head/discharge relation). Then Q depends on only one measured quantity – the upstream head H. Examples: weirs (thin-plate, broad-crested), flumes (Venturi, Parshall, long-throated). Their advantage is high accuracy (± 2–5 %) and simplicity; disadvantage is head loss and possible sediment deposition.

★ **Key Point:** A measuring structure must satisfy three conditions: (i) a well-defined control (critical flow or free nappe), (ii) a calm, straight approach channel with uniform velocity distribution, and (iii) head measured at the correct location (upstream, about 3–4 H_{max} from the weir, in a stilling well if needed).

2. SHARP-CRESTED WEIRS

2.1 Description

A **sharp-crested (thin-plate) weir** is a vertical plate with a sharp upstream edge (bevelled 45° downstream, crest thickness ≤ 2 mm) placed across the channel normal to the flow. Water passes over the crest in a **nappe** that springs free from the plate. The weir may be rectangular, triangular (V-notch), trapezoidal (Cipolletti) or proportional (Sutro).

Requirements: crest level above the bed (weir height P) ≥ $2H_{max}$ and ≥ 0.3 m; head H ≥ 0.06 m (surface-tension effects are important at smaller head); H ≤ 0.6 m; channel width ≥ 2 H; the nappe

must be **fully aerated** (vented), downstream water level below crest (free flow).

2.2 Rectangular weir – derivation

Consider a horizontal strip of thickness dh at depth h below the free surface, width L. Velocity at the strip $\sqrt{(2gh)}$. Discharge through the strip: $dQ = C_d\, L\, \sqrt{(2gh)}\, dh$. Integrating from 0 to H:

$$\mathbf{Q = (2/3)\, C_d\, L\, \sqrt{(2g)}\, H^{3/2}}$$

This is the basic formula. Typical $C_d = 0.60–0.62$. Rehbock's formula: $C_d = 0.602 + 0.083\, H/P$ (equivalently $Q = 1.78\, L\, H^{3/2}(1 + 0.14\, H/P)$ in SI).

End contractions: if the weir does not span the full channel width, the nappe contracts at the sides and the effective length is reduced (**Francis formula**):

$$\mathbf{Q = 1.84\, (L - 0.1\, n\, H)\, H^{3/2}}$$ **(SI; n = number of end contractions: 2 for fully contracted, 0 for suppressed weir)**

Velocity of approach: when $V_a = Q/(B(P + H))$ is not negligible, use the total head $H + h_a$, $h_a = V_a^2/2g$:

$$\mathbf{Q = 1.84\, L\, [\, (H + h_a)^{3/2} - h_a^{3/2}\,]}$$ **(suppressed weir)**

2.3 Triangular (V-notch) weir

Used for **small discharges** (< 30 L/s) because a small change in Q produces a large change in H, so it is more sensitive than the rectangular weir. For a notch of angle θ:

$$\mathbf{Q = (8/15)\, C_d\, \sqrt{(2g)}\, \tan(\theta/2)\, H^{5/2}}$$

For a 90° notch ($C_d = 0.58$): $\mathbf{Q \approx 1.38\, H^{5/2}}$ (SI). For 60° notch: $Q \approx 0.80\, H^{5/2}$. Range of H: 0.06–0.38 m.

2.4 Trapezoidal (Cipolletti) weir

Side slopes 1 horizontal : 4 vertical. The extra area at the sides compensates for the end contraction loss, so the discharge is the same as the suppressed rectangular weir of the same crest length:

$$\mathbf{Q = 1.86\, L\, H^{3/2}}$$ **(SI)**

2.6 Submerged (drowned) weir

When the tailwater rises above the crest, the discharge reduces. **Villemonte (1947):**

$$\mathbf{Q_s = Q_1\, [\, 1 - (H_2/H_1)^n\,]^{0.385}}$$

where Q_1 is the free-flow discharge for the upstream head H_1, H_2 = downstream head over the crest, and n = 1.5 (rectangular), 2.5 (V-notch).

2.7 Worked example

A rectangular, fully contracted sharp-crested weir has L = 1.2 m and a measured head H = 0.25 m. Find the discharge using the Francis formula.

- $L' = L - 0.1\, n\, H = 1.2 - 0.1 \times 2 \times 0.25 = 1.15$ m.
- $H^{3/2} = 0.25^{1.5} = 0.125$.

- Q = 1.84 × 1.15 × 0.125 = **0.264 m^3/s**.

For a 90° V-notch with H = 0.20 m: Q = 1.38 × $0.20^{2.5}$ = 1.38 × 0.0179 = 0.0247 m^3/s ≈ 25 L/s.

2.8 Errors in measurement

Error in Q due to error in H: for rectangular weir dQ/Q = 1.5 dH/H; for V-notch dQ/Q = 2.5 dH/H. A 1 % error in H gives a 1.5 % (rectangular) or 2.5 % (triangular) error in Q. Therefore the head must be measured carefully (point gauge or float in stilling well, accuracy ± 0.3 mm).

✎ **Exam Question:** Derive the discharge formula for a V-notch weir. Why is it preferred to a rectangular weir for small discharges? (8 marks)

3. BROAD-CRESTED WEIRS

3.1 Description

A **broad-crested weir** is a raised, flat-topped obstruction across the channel whose crest length L in the flow direction is large enough for the streamlines on the crest to be parallel and the pressure hydrostatic. Flow passes through critical depth on the crest, which is the control section. The upstream face is rounded or bevelled to avoid separation.

Limits of application: 0.08 < H_1/L < 0.5 (H_1 = upstream head above the crest). If H_1/L < 0.08, boundary friction dominates; if H_1/L > 0.5 the weir behaves as a **short-crested weir** (curvature of streamlines is significant). Typical L = 0.5–3 m for field weirs.

3.2 Theory

Consider a rectangular weir and a free flow (no tailwater influence). Let H be the upstream head above the crest (measured at ≥ 3H upstream). Neglect the velocity of approach and losses. Applying Bernoulli between the upstream section and the crest:

H = $y_c + V_c^2/2g$. For critical flow $V_c^2/2g = y_c/2$, so

y_c = (2/3) H and $V_c = \sqrt{(g\, y_c)} = \sqrt{(2gH/3)}$

$q = y_c\, V_c$ = (2/3) H $\sqrt{(2gH/3)}$:

$$\mathbf{Q = (2/3)^{3/2}\ \sqrt{g}\ L\ H^{3/2} = 1.705\ L\ H^{3/2}}$$

Including a **discharge coefficient** C_d (for friction and curvature) and the velocity-of-approach coefficient C_v:

$$\mathbf{Q = 1.705\ C_d\ C_v\ L\ H^{3/2}\ (SI)}$$

with C_d ≈ 0.85–1.0 for well-built weirs (Bos recommends C_d = 0.93 + 0.10 H/L for 0.1 < H/L < 0.7, approximately). The discharge is proportional to $H^{3/2}$ and **independent of tailwater** provided the weir is **modular**.

3.3 Modular limit

Free (modular) flow exists when the critical depth forms on the crest; the tailwater level (head H_2 above crest) must not exceed a limit:

H_2/H_1 ≤ 0.66–0.75 (modular limit). For larger submergence the flow is **drowned** and Q is reduced; correction for submergence is required (Tailwater correction factor, or two-head measurement).

3.5 Comparison: sharp-crested vs broad-crested

Feature	Sharp-crested	Broad-crested
Accuracy	High (± 1–3 %)	Moderate (± 3–5 %)
Head loss	Large	Small to moderate
Sediment / debris	Collects sediment (poor)	Passes sediment better
Strength	Fragile (plate)	Robust (concrete)
Q–H relation	$Q \propto H^{3/2}$ (rect.), $H^{5/2}$ (V)	$Q \propto H^{3/2}$
Control section	Free nappe	Critical depth on crest
Typical use	Laboratory, small canals	Field channels, irrigation

3.6 Worked example

A broad-crested weir (L_{crest} = 2 m in the flow direction, width b = 5 m) is built in a canal. Head measured above the crest H = 0.60 m. Take C_d = 0.92 and C_v = 1.0.

- Check H/L = 0.30 (within 0.08–0.5 ✓).
- $Q = 1.705 \times 0.92 \times 5 \times 0.60^{1.5} = 1.705 \times 0.92 \times 5 \times 0.4648 =$ **3.65 m^3/s**.
- y_c = 0.40 m; V_c = 1.98 m/s; the downstream water depth above the crest must be < 0.66 × 0.60 = 0.40 m for modular flow.

✎ **Exam Question:** Derive the discharge equation for a broad-crested weir and explain the concept of modular limit. (10 marks)

4. CRITICAL-DEPTH FLUMES

4.1 Principle

A **flume** is a specially shaped channel section in which the width is contracted and/or the bed raised so that the flow passes through critical depth in the throat. Compared to weirs, flumes cause **less head loss**, pass sediment and debris, and need no stilling pool. Because critical flow prevails in the throat, the discharge is a unique function of the upstream head (provided submergence is below the modular limit).

4.2 Venturi flume (standing-wave flume)

Contraction in width (and sometimes bed hump). For a rectangular throat of width b_t with critical depth $y_c = (2/3)H$:

$$\mathbf{Q = C_d\,(2/3)^{3/2}\,\sqrt{g}\,b_t\,H^{3/2} = 1.705\,C_d\,b_t\,H^{3/2}}$$

Here H = upstream total energy head above the throat bed. $C_d \approx 0.95$–0.99 (affected by boundary-layer growth). A gradual converging section (typically 1:3 to 1:5) minimises energy loss; a diverging section (1:6) recovers some head.

4.3 Parshall flume

Developed by R. L. Parshall (1920s, Colorado). Consists of a **converging section**, a **throat** with a downward sloping floor and a **diverging section** with an upward sloping floor. Size is designated by the throat width W (from 1 in to 50 ft). Two stilling wells measure heads: H_a (in converging section at 2/3 of its length) and H_b (at the throat).

Free-flow discharge (US units: W and H_a in feet, Q in ft^3/s):

$$Q = 4\ W\ H_a^{1.522\ W^{0.026}}$$

Example: W = 1 ft, H_a = 0.5 ft: Q = 4 × $0.5^{1.522}$ = 1.39 ft^3/s ≈ 39 L/s. (SI rating tables are available for every standard size.) Submergence S = H_b/H_a; free flow if S is below the transition limit: **50 % (W = 1–3 in), 60 % (6–9 in), 70 % (1–8 ft), 80 % (10–50 ft)**. For higher submergence a correction (from charts) is applied and both H_a and H_b are needed.

Advantages: works with free and submerged flow; small head loss (about 1/4 of a weir); self-cleaning; insensitive to approach velocity; wide range of sizes (1 to 1000s L/s).
Disadvantages: complicated geometry, needs precise construction; high cost; relatively sensitive to changes from standard dimensions.

4.5 Long-throated flumes (Bos, Clemmens, Replogle)

A flume with a throat long enough (L/H > 1–3) that the streamlines are parallel and hydrostatic and critical depth forms in the throat. The **rating can be computed theoretically** for any shape (rectangular, trapezoidal, circular, U-shape) with the energy equation and boundary-layer correction, so no laboratory calibration is needed:

$$Q = C_d\ A_c\ \sqrt{(g\ A_c/T_c)},\ H_1 = y_c + A_c/(2T_c)\ (A_c,\ T_c \text{ at critical depth})$$

The program **WinFlume** (USBR/ILRI) designs the flume and the rating table. These flumes are standard for irrigation turnouts.

4.6 Choice of measuring structure

Condition	Preferred device
Very small flows (< 30 L/s), accurate lab work	90° or 60° V-notch weir
Medium flows, clean water	Rectangular sharp-crested weir
Sediment-laden canals, small head available	Flume (Parshall, cutthroat, long-throated)
Field irrigation canals, robust, low maintenance	Broad-crested or Crump weir; RBC flume
Wide variation in flow and unknown submergence	Parshall flume

★ **Key Point:** Modular limit rule: weirs and flumes work best when the downstream water level is low enough that flow is "free". Always check submergence under the maximum flow.

5. RECENT ADVANCEMENTS IN OPEN-CHANNEL FLOW MEASUREMENT

5.1 Limitations of conventional methods

Structures need head loss and maintenance; current-meter gauging is slow, labour-intensive, risky in floods and cannot give continuous records. Modern technologies provide non-intrusive, rapid, automated and remote measurements.

5.2 Acoustic Doppler technologies

- **Acoustic Doppler Current Profiler (ADCP):** transmits sound pulses along 3–4 beams and measures the Doppler shift of echoes from suspended particles to give velocity profiles in cells through the depth. Moving-boat ADCP (e.g. on a tethered or remote-controlled boat) measures a whole cross-section in 5–15 minutes, including bottom-tracking to measure boat velocity and depth; Q = ∫∫ v·n dA. Used widely by USGS, CWC etc. Fixed (horizontal) H-ADCP mounted on the bank gives **index-velocity** rating for continuous discharge.

- **Acoustic Doppler Velocimeter (ADV):** high-frequency point measurement of 3-D velocity (to 200 Hz) in laboratory and field; gives turbulence characteristics and Reynolds stresses.
- **Ultrasonic transit-time (acoustic path) flow meters:** sound pulses are sent in both directions across the channel along one or more paths; the time difference gives the path-average velocity; used in power canals, large rivers and tidal channels.
- **Ultrasonic level sensor + weir/flume:** non-contact level measurement for automatic head recording.

5.3 Radar and optical (non-contact) methods

- **Surface-velocity radar:** Doppler radar (K- or X-band) mounted on a bridge or a bank measures the surface velocity; the mean velocity $V = \alpha V_s$ ($\alpha \approx 0.85$) is used with surveyed cross-section to give Q. Safe in floods.
- **Radar level sensors** for stage.
- **Image-based methods: LSPIV** (large-scale particle image velocimetry), **PTV** and **STIV** (space-time image velocimetry) track natural surface features (foam, ripples) or seeding in video from fixed cameras or **UAV/drones**, giving 2-D surface velocity fields. Low cost and no contact; calibrated by ground control points.
- **Satellite and UAV remote sensing:** width and water-surface slope from satellite altimetry (e.g. SWOT), Landsat/Sentinel imagery for river width; Q estimated from at-many-stations hydraulic geometry (AMHG).

5.5 Digital and smart-hydrology developments

- **IoT telemetry:** loggers send stage, velocity and discharge in real time via GSM/satellite/LoRa to SCADA systems; used for canal automation and flood warning.
- **Machine learning and data assimilation:** ANN, random forests and LSTM networks to derive rating curves, fill gaps, and estimate discharge from camera images.
- **Smart/automatic gates (Constant-downstream-level, Baffle modules)** that measure and control flow simultaneously (e.g. flume gates and constant-downstream-level gates).
- **Standards:** ISO 748 (velocity–area), ISO 1438 (thin-plate weirs), ISO 4359 (flumes), ISO 6416 (ultrasonic velocity method), ISO 24578 (acoustic Doppler profilers).

5.6 Comparison of modern techniques

Technique	Advantages	Limitations
Moving-boat ADCP	Fast, whole cross-section, safe	Bed-load error, expensive, needs minimum depth
H-ADCP / index-velocity	Continuous Q, backwater ok	Needs calibration
Surface-velocity radar	Non-contact, safe in floods	Surface only; wind effects
LSPIV / drone	Low cost, 2-D flow field	Needs seeding/features, lighting
Ultrasonic transit time	Accurate (± 1–2 %)	Needs stable channel, installation cost
Tracer dilution	No cross-section needed	Needs good mixing length

✎ **Exam Question:** Discuss recent advancements in open-channel flow measurement and compare ADCP and radar methods. (10 marks)

6. DIMENSIONAL ANALYSIS

6.1 Purpose

Dimensional analysis expresses a physical relation in terms of **dimensionless groups**. It reduces the number of variables, guides experiments, and forms the basis for model–prototype similarity. Fundamental dimensions: mass **M**, length **L**, time **T** (and temperature if needed).

6.3 Rayleigh's method

Write the dependent variable as a product of powers of independent variables, then equate indices of M, L, T. It is suitable for ≤ 4 variables. Example: discharge over a rectangular weir $Q = K\,L^a\,H^b\,g^c$ → dimensions $L^3T^{-1} = L^{a+b+c}\,T^{-2c}$ → c = 1/2 and a + b = 5/2. Since discharge is proportional to the width, a = 1 and b = 3/2, giving $\mathbf{Q = K\,L\,H^{3/2}\,g^{1/2}}$, the same form as the weir formula.

6.4 Buckingham π theorem

If a physical phenomenon involves n variables and m fundamental dimensions, it can be expressed as a relationship between **(n − m) independent dimensionless π terms**. Procedure:

- 1. List all variables (n) affecting the phenomenon, with their dimensions.
- 2. Count fundamental dimensions (m); number of π terms = n − m.
- 3. Select m **repeating variables** that (a) together contain all fundamental dimensions, (b) cannot form a dimensionless group among themselves – commonly one geometric (D or L), one kinematic (V) and one dynamic (ρ).
- 4. Form each π by multiplying one non-repeating variable by the repeating variables with unknown exponents, and solve exponents so that the product is dimensionless.
- 5. Express the result as $\pi_1 = \varphi(\pi_2, \pi_3, \ldots)$. Combine groups if needed (product/ratio) to get standard numbers.

Example – pressure drop in a pipe. Δp = f(ρ, V, D, μ, L, ε): n = 7, m = 3, so 4 π-terms. Choose repeating variables ρ, V, D:

- $\pi_1 = \Delta p/(\rho V^2)$ (Euler number)
- $\pi_2 = \mu/(\rho VD) = 1/Re$
- $\pi_3 = L/D$, $\pi_4 = \varepsilon/D$

So $\Delta p/(\rho V^2) = \varphi(Re, L/D, \varepsilon/D)$ → $h_f = f\,(L/D)(V^2/2g)$ – the Darcy–Weisbach equation, with $f = \varphi(Re, \varepsilon/D)$.

6.5 Important dimensionless numbers

Number	Expression	Ratio of forces	Governs
Reynolds	$Re = \rho VL/\mu$	Inertia : viscous	Pipe flow, boundary layer, flow around bodies
Froude	$Fr = V/\sqrt{(gL)}$	Inertia : gravity	Open-channel flow, spillways, waves, ship resistance
Euler	$Eu = \Delta p/(\rho V^2)$	Pressure : inertia	Pressure drops, cavitation ($\sigma = (p - p_v)/(0.5\rho V^2)$)
Weber	$We = \rho V^2 L/\sigma$	Inertia : surface tension	Small-scale free-surface flow, droplets, thin nappes
Mach / Cauchy	Ma = V/c	Inertia : elastic	Compressible flow, water hammer
Strouhal	St = fL/V	Unsteady : inertia	Vortex shedding

7. PHYSICAL MODELLING IN HYDRAULICS

7.1 Why physical models?

Many hydraulic problems are too complex for theory alone: 3-D flow around piers and intakes, energy dissipators, spillways, harbours, river training, sediment transport, cavitation and air entrainment. A **physical (scale) model** is a reduced (or sometimes enlarged) replica built to reproduce the flow, in which measurements are made and then converted to prototype values. Numerical (CFD) models are increasingly used, but a physical model remains the most reliable for complex free-surface flow and for validating numerical codes.

Terminology: **prototype** = the actual structure; **model** = replica; **scale ratio** (X_r) = prototype value / model value (e.g. $L_r = L_p/L_m$). In these notes we use model/prototype ratio $X_r = X_m/X_p$; both conventions occur, so check the question.

7.2 Similitude (similarity) conditions

- **Geometric similarity:** all linear dimensions in a constant ratio L_r; areas $A_r = L_r^2$; volumes L_r^3. Includes roughness, i.e. $\varepsilon_r = L_r$.
- **Kinematic similarity:** velocities and accelerations at corresponding points are in a constant ratio (streamline patterns are similar).
- **Dynamic similarity:** ratio of all forces (inertia, gravity, viscous, pressure, surface tension, elastic) is the same in model and prototype. If dynamic similarity is satisfied with geometric similarity, kinematic similarity follows.

Dynamic similarity requires equality of the relevant dimensionless numbers. Since it is impossible to equate Re, Fr and We simultaneously with the same fluid, the **dominant force** is identified and the corresponding number is equated; other forces are controlled by choosing model size so that their effects are negligible (scale effects).

8. MODELLING OF CLOSED-CONDUIT FLOWS AND FREE-SURFACE FLOWS

8.1 Closed-conduit (pressurised) flows – Reynolds model law

In pressure flows (pipes, valves, turbines, pumps, bends), viscous and inertia forces dominate while gravity only affects the hydrostatic pressure, which does not influence the flow pattern. Hence equality of the **Reynolds number**:

$$(\rho VL/\mu)_m = (\rho VL/\mu)_p \rightarrow V_r\, L_r/\nu_r = 1$$

Scale ratios (Reynolds model, subscript r = model/prototype):

Quantity	Ratio	Same fluid ($\nu_r = 1$)
Length	L_r	L_r
Velocity	$V_r = \nu_r/L_r$	$1/L_r$
Time	$T_r = L_r^2/\nu_r$	L_r^2
Discharge	$Q_r = \nu_r\, L_r$	L_r
Force	$F_r = \rho_r\, \nu_r^2$	1
Pressure	$p_r = \rho_r\, \nu_r^2/L_r^2$	$1/L_r^2$

Practical problem: with the same fluid (water) in a small model, the velocity would need to be larger by $1/L_r$, which is often impractically high (e.g. a 1:10 model needs 10 × prototype velocity).

Remedies: use a different fluid (e.g. air, oil with lower ν), pressurise or heat the water (reduces ν), or rely on **Reynolds-number independence** – above a critical Re (fully rough turbulent flow) friction factor is independent of Re, so only geometric (including roughness) and a "sufficiently large" Re are required. For pipe flow it is enough that the model Re > 10^4.

Cavitation models also equate the cavitation number $\sigma = (p - p_v)/(0.5\rho V^2)$. **Water-hammer models** use the Cauchy (Mach) number – scale a, V and time with L/a.

8.2 Free-surface flows – Froude model law

In open channels, spillways, flow over weirs, river and harbour flows, gravity is the governing force, and viscous effects are secondary. Therefore **Froude number** must be equal:

$$\mathbf{(V/\sqrt{(gL)})_m = (V/\sqrt{(gL)})_p \rightarrow V_r = \sqrt{(L_r)}\ (g_r = 1)}$$

Scale ratios (Froude model):

Quantity	Ratio	Exponent of L_r
Length	L_r	1
Area	A_r	2
Volume		3
Velocity	V_r	1/2
Time	$T_r = L_r/V_r$	1/2
Acceleration		0
Discharge	$Q_r = A_r V_r$	5/2
Force	$F_r = \rho_r L_r^3$	3
Pressure	$p_r = \rho_r L_r$	1
Power	$P_r = F_r V_r$	7/2
Manning's n	$n_r = L_r^{1/6}$	1/6
Kinematic viscosity (if scaled)	$\nu_r = L_r^{3/2}$	3/2

Worked example: A spillway has a design discharge of 4500 m^3/s. A Froude model of scale 1:50 (L_r = 1/50) is built.

- $Q_m = 4500 \times (1/50)^{5/2} = 4500/17\,678 =$ **0.255 m^3/s**.
- A prototype velocity of 20 m/s corresponds to $V_m = 20/\sqrt{50} = 2.83$ m/s.
- A prototype flood of 12 h lasts $12/\sqrt{50} = 1.70$ h in the model.
- A measured model force of 20 N corresponds to a prototype force = $20 \times 50^3 = 2.5$ MN ($\rho_r = 1$).
- A measured head of 0.12 m in the model = 6.0 m in the prototype.

8.3 Scale effects

Because water is used in both model and prototype, **viscosity and surface tension are not scaled** correctly in a Froude model. The model Reynolds number is lower than prototype by $L_r^{3/2}$, so friction, boundary-layer thickness, air entrainment and wave breaking are not reproduced exactly. To limit scale effects:

- Choose a model large enough so that **flow depth ≥ 3 cm**, **head over the crest ≥ 5 cm** (We effect), **Re ≥ 2000–5000** (turbulent flow in model), and model channel R ≥ ~1–2 cm.
- Adjust roughness so that the Froude-scaled $n_r = L_r^{1/6}$ is obtained.
- For **air entrainment**, model scales larger than 1:15 are preferred; extrapolate results if needed.
- **Cavitation** cannot be modelled by a Froude model – use a separate, low-pressure test.

★ **Key Point:** Typical Froude-model scales: spillways and energy dissipators 1:20 to 1:60; intake/vortex studies 1:5 to 1:25; harbours 1:100 to 1:150 horizontally; rivers 1:100 to 1:500 horizontally (distorted).

✎ **Exam Question:** A 1:25 Froude model of a spillway has a discharge of 0.5 m^3/s. Find the prototype discharge and the ratio of the forces. *Ans:* $Q_p = 0.5 \times 25^{5/2} = 0.5 \times 3125 = 1562\ m^3/s$; $F_p/F_m = 25^3 = 15\ 625$.

9. DISTORTED MODELS

9.1 What is a distorted model?

A **distorted model** is one in which the **horizontal scale X_r and vertical scale Y_r are different**. The **distortion ratio** (distortion factor) is

$n_d = Y_r/X_r$ (vertical scale ratio ÷ horizontal scale ratio)

with scale ratios taken as model/prototype. Example: horizontal 1:400 and vertical 1:40 give n_d = (1/40)/(1/400) = 10.

9.2 Why distort?

- Large rivers/estuaries/harbours have lengths of kilometres and depths of metres. An undistorted model at 1:1000 would have depths of millimetres: surface tension and laminar flow, very high viscous effects, and measuring accuracy is lost.
- Need to reproduce **sufficient bed shear to move sediment**, and keep **turbulent flow (Re > 2000)**.
- Fit the model within the available laboratory space.
- Get enough **slope** for gravity-driven flow in the model.

9.3 Scale relationships for distorted Froude models

Let X_r = horizontal scale ratio and Y_r = vertical scale ratio (model/prototype). In a distorted model $Y_r > X_r$, i.e. the vertical is exaggerated.

Quantity	Relation	Derivation
Velocity	$V_r = \sqrt{(Y_r)}$	Froude number (depth as length scale)
Discharge	$Q_r = X_r Y_r^{3/2}$	Q = width × depth × velocity
Time (horizontal movement)	$T_r = X_r/\sqrt{Y_r}$	t = distance/velocity
Slope	$S_r = Y_r/X_r$	vertical ÷ horizontal
Hydraulic radius	$R_r \approx Y_r$ (wide channel)	R ≈ y
Manning's n	$n_r = Y_r^{2/3}/X_r^{1/2}$	$V = R^{2/3}S^{1/2}/n$
Time (vertical movement, e.g. settling)	$Y_r/V_{r,v}$	Depends on fall velocity

Example: A river model has X_r = 1/400 and Y_r = 1/40. Distortion = 10. $Q_r = (1/400)(1/40)^{3/2}$ = 1/(400 × 253) = 1/101 192. So a prototype flood of 5000 m^3/s corresponds to 0.049 m^3/s. V_r = 1/6.32 (a prototype 3 m/s becomes 0.47 m/s). T_r = (1/400)/(1/6.32) = 1/63.2: a 24 h prototype event lasts 22.8 min. Slope in model = 10 × prototype slope. $n_r = (1/40)^{2/3}/(1/400)^{1/2}$ = 0.0852 × 20 = 1.70, so the model roughness must be 1.7 times the prototype n – use artificial roughness (rods, wire mesh, gravel) if the smooth model bed is not rough enough; if less roughness is needed,

polish the bed.

9.4 Limitations of distortion

- Velocity distribution and secondary currents (bends) are different; **3-D effects** and eddies are not scaled correctly.
- Side-slope and bank angles are changed: steeper banks → different stability and scour.
- **Wave patterns** and refraction in harbours are affected; vertical accelerations become important; hydrostatic assumption can fail.
- Sediment transport: grain size, fall velocity and shear stress must be re-scaled for the movable bed; time scales for bed-load and suspended load are different.
- Limit distortion to about 5–10 (maximum ≈ 20) for rivers; use undistorted models for local structures (bridge piers, intake).

★ **Key Point:** Rule of thumb: distort only the minimum necessary; avoid distortion where flow patterns (separation, eddies, local scour) are important.

10. DESIGN OF PHYSICAL MODELS

10.1 Step-by-step procedure

- **Step 1 – Define objectives.** What will the model show: discharge capacity, pressures, water-surface profile, scour, energy dissipation, forces, sediment movement? Choose the dominant forces (gravity → Froude; viscosity → Reynolds; surface tension → Weber).
- **Step 2 – Select the type of model.** Rigid or movable bed; undistorted or distorted; sectional or comprehensive; fixed or tidal.
- **Step 3 – Select the scale.** Compromise between (a) space and discharge available in the laboratory, (b) cost and construction time, (c) measurement accuracy, (d) scale-effect limits (minimum depth ≈ 3 cm; weir head ≥ 5 cm; Re ≥ 2000; We effects small).
- **Step 4 – Derive scale ratios** for all variables (Section 8, 9), and calculate model discharge, velocities, depths, slopes, roughness, time scale.
- **Step 5 – Layout and construction.** Provide an approach channel (≥ 5–10 channel widths, with flow straighteners/baffles), a tail-gate to control the downstream level, a stilling tank and calibrated flow meter. Materials: concrete, marine plywood, acrylic (transparent, for flow visualisation), PVC, 3-D printed parts, cement mortar finishes (set to the roughness needed).
- **Step 6 – Instrumentation:** point and hook gauges (water level), Pitot tubes, ADV, propeller meters, pressure transducers/piezometers, load cells, PIV/LSPIV, dye tracers, high-speed video.
- **Step 7 – Verification (calibration):** reproduce a known prototype condition (rating curve, high-water marks, tide records) by adjusting roughness; then test the proposed design for different alternatives.
- **Step 8 – Testing and interpretation:** conduct runs, record data, convert to prototype units, apply scale-effect corrections, and provide recommendations.

10.2 Movable-bed models

The bed material must be scaled so the model simulates **incipient motion, bed forms and transport**. Conditions: equal Shields (mobility) parameter, equal Froude number, particle Reynolds number R_* above ~ 10–70. Because natural sand scaled by 1/100 would become cohesive silt, **lightweight materials** (crushed coal, bakelite, polystyrene, walnut shells, plastic pellets) are used, density $\rho_{s,m}$ ~ 1050–1500 kg/m^3. The time scale of bed changes is found by sediment-continuity (Exner) scaling and usually differs from the flow time scale (an extra **sediment time-scale factor**).

10.3 Example – design of a spillway model

Prototype: ogee spillway, crest length 60 m, design head 8 m, design flood 3000 m^3/s; laboratory flume width 1.2 m and maximum flow 0.25 m^3/s. Choose Froude scale.

- Discharge limit: for the full 60 m crest, $L_r = 1/40$ would need $Q_m = 3000/40^{2.5} = 0.296\ m^3/s >$ 0.25 and a 1.5 m wide model > 1.2 m flume. So build a **sectional model** of a 48 m strip of crest (1.2 m in the flume).
- Unit discharge: $q_p = 3000/60 = 50\ m^2/s$; $q_m = 50/40^{1.5} = 0.198\ m^2/s$; $Q_m = 0.198 \times 1.2 = 0.237\ m^3/s \le 0.25$ ✓.
- Model design head = 8/40 = 0.20 m (> 0.05 m ✓, so surface-tension effects are negligible).
- Time scale: $1/\sqrt{40} = 1/6.3$. Velocity scale 1/6.3.

10.5 Quick revision – formulae

Topic	Formula
Rectangular sharp-crested weir	$Q = (2/3)C_d\sqrt{(2g)}\ L\ H^{3/2}$
Francis	$Q = 1.84(L - 0.1nH)H^{3/2}$
V-notch	$Q = (8/15)C_d\sqrt{(2g)}\ \tan(\theta/2)\ H^{5/2}$
Broad-crested weir	$Q = 1.705\ C_d\ L\ H^{3/2}$
Critical depth on crest	$y_c = (2/3)H$
Parshall flume (ft, cfs)	$Q = 4\ W\ H_a^{1.522\ W^{0.026}}$
Froude model	$V_r = L_r^{1/2}$, $Q_r = L_r^{5/2}$, $T_r = L_r^{1/2}$, $F_r = L_r^3$
Reynolds model	$V_r = \nu_r/L_r$, $Q_r = \nu_r L_r$
Distorted model	$V_r = \sqrt{Y_r}$, $Q_r = X_r Y_r^{3/2}$, $T_r = X_r/\sqrt{Y_r}$

✎ **Exam Question:** Explain with suitable examples why distorted models are used in hydraulics. Derive the scale ratios for velocity, discharge and time for a distorted model. (10 marks)